

Sheet 6

1.a)

Let $P_{data}: \Omega \rightarrow X$, $P_g: \Omega \rightarrow X$ be random variables.

$$\begin{aligned}
 V(G, D) &= E_{x \sim P_{data}} [\log D(x)] + E_{z \sim P_z} [\log(1 - D(G(z)))] \\
 &= E_{x \sim P_{data}} [\log D(x)] + E_{x \sim P_g} [\log(1 - D(x))] \\
 &= E_{x \sim P_{data}} [\log D(x)] + E_{x \sim P_g} [\log(1 - D(x))] \\
 &= \sum_{x \in X} P_{data}(x) \log D(x) + P_g(x) \log(1 - D(x)) \\
 &= a \log \gamma + b \log(1 - \gamma) \xrightarrow{\text{hint}} D^* = \frac{P_{data}(x)}{P_{data}(x) + P_g(x)} \\
 &\quad \begin{matrix} a = P_{data}(x) \\ b = P_g(x) \\ \gamma = D(x) \end{matrix}
 \end{aligned}$$

1.b)

$$\begin{aligned}
 C(G) &= \max_D V(G, D) = V(G, D^*) \\
 &= E_{x \sim P_{data}} [\log D^*(x)] + E_{z \sim P_z} [\log(1 - D^*(G(z)))] \\
 &\stackrel{P_g = G(z)}{\downarrow} = E_{x \sim P_{data}} [\log D^*(x)] + E_{x \sim P_g} [\log(1 - D^*(x))] \\
 &= E_{x \sim P_{data}} \left[\log \left(\frac{P_{data}(x)}{P_{data}(x) + P_g(x)} \right) \right] + E_{x \sim P_g} \left[\log \left(1 - \frac{P_{data}(x)}{P_{data}(x) + P_g(x)} \right) \right] \\
 &= E_{x \sim P_{data}} \left[\log \left(\frac{P_{data}(x)}{P_{data}(x) + P_g(x)} \right) \right] + E_{x \sim P_g} \left[\log \left(\frac{P_g(x)}{P_{data}(x) + P_g(x)} \right) \right]
 \end{aligned}$$

1.c)

$$\begin{aligned}
 M(x) &= \frac{1}{2} P_{data}(x) + \frac{1}{2} P_g(x) \\
 E_{x \sim P_{data}} [\log D(x)] &= E_{x \sim P_{data}} \left[\log \left(\frac{P_{data}(x)}{\frac{P_{data}(x) + P_g(x)}{2}} \right) \right] - \log 2 \\
 &= E_{x \sim P_{data}} \left[\log \left(\frac{P_{data}(x)}{M(x)} \right) \right] - \log 2 = KL(P_{data} \parallel M) - \log 2 \\
 E_{z \sim P_z} [\log(1 - D(G(z)))] &= E_{z \sim P_z} \left[\log(1 - D(\gamma)) \mid \gamma = G(z) \right] \\
 &= E_{z \sim P_z} \left[\log \left(1 - \frac{P_{data}(\gamma)}{P_{data}(\gamma) + P_g(\gamma)} \right) \right] = E_{z \sim P_z} \left[\log \left(\frac{P_g(\gamma)}{P_{data}(\gamma) + P_g(\gamma)} \right) \right] \\
 &= E_{z \sim P_z} \left[\log \left(\frac{P_g(\gamma)}{\frac{P_{data}(\gamma) + P_g(\gamma)}{2}} \right) \right] - \log 2 \\
 &= E_{z \sim P_z} \left[\log \left(\frac{P_g(\gamma)}{\frac{P_{data}(\gamma) + P_g(\gamma)}{2}} \right) \right] - \log 2 = KL(P_g \parallel M) - \log 2 \\
 \min_G C(G) &= \min_G KL(P_{data} \parallel M) + KL(P_g \parallel M) - \log 4 \\
 &= \min_G \underbrace{2 JSD(P_{data} \parallel P_g)}_{0 \leq} - \log 4 \\
 \text{Let } G^* \text{ be s.t. } P_g &= P_{data}, \text{ then } JSD(P_{data} \parallel P_g) = 0 \\
 &= -\log 4
 \end{aligned}$$

2.

$$\begin{aligned}
 P_\theta(x_{0:T}) &= P(x_T) \prod_{t=1}^T P(x_{t-1} | x_t) \\
 q(x_{1:T} | x_0) &= \prod_{t=1}^T q(x_t | x_{t-1}, x_0) \\
 &= q(x_1 | x_0) \prod_{t=2}^T \frac{q(x_{t-1} | x_t, x_0) q(x_t, x_0)}{q(x_{t-1}, x_0)} \\
 &= q(x_T) \prod_{t=2}^T q(x_{t-1} | x_t, x_0) \\
 -\log \frac{P_\theta(x_{0:T})}{q(x_{1:T} | x_0)} &= \log \frac{q(x_{1:T} | x_0)}{P_\theta(x_{0:T})} \\
 &= \log \frac{q(x_T) \prod_{t=2}^T q(x_{t-1} | x_t, x_0)}{P(x_T) P(x_0 | x_1) \prod_{t=2}^T P(x_{t-1} | x_t)} \\
 &= -\log P(x_0 | x_1) + \sum_{t=2}^T \frac{q(x_{t-1} | x_t, x_0)}{P(x_{t-1} | x_t)} + \log \frac{q(x_T)}{P(x_T)} \\
 E_q \left[-\log \frac{P_\theta(x_{0:T})}{q(x_{1:T} | x_0)} \right] &= \mathcal{L}_0 + \mathcal{L}_1 + \dots + \mathcal{L}_n \\
 \mathcal{L}_0 &= -E_q [P(x_0 | x_1)] \\
 \mathcal{L}_{t-1} &= E_q \left[\sum_{t=2}^T \frac{q(x_{t-1} | x_t, x_0)}{P(x_{t-1} | x_t)} \right] \\
 &= KL(q(x_{t-1} | x_t, x_0) \parallel P(x_{t-1} | x_t)) \\
 \mathcal{L}_T &= E_q \left[\log \frac{q(x_T)}{P(x_T)} \right] = KL(q(x_T) \parallel P(x_T))
 \end{aligned}$$